

CS1001- Introduction to Computing (ITC)

Introduction to course

Who am I?

- My research interests lie in tackling intriguing and challenging interdisciplinary problems within the fields of image processing, medical image analysis, deep learning, computer vision, image compression, signal processing, **Generative AI, and Explainable AI.**

Teaching Experience: Almost 10 years

- Former Assistant Professor, PSIT Kanpur, 2015-2017
- Lecturer, PSIT Kanpur, 2010-2013

Research Experience: Almost 9 years (including PhD and MU)

- Worked on one Computer Vision Project at IIT Kanpur in 2015

4-Credit course (3-0-2)

✓ Lectures: 3 hours

✓ Lab: 2 hours

✓ Class Timings:

- Monday: 12:35 pm – 1:30 pm, ECR-4
- Tuesday: 11:35 am – 12:30 pm, ECR-4
- Thursday: 11:35 am – 12:30 pm, ECR-4

✓ Lab Timings:

- Monday: 4:35 – 06:30 pm, Main Block, First Floor, Computer Lab 1 and 2

Course methodology

- You will be given a lot of reading resources for each class/lab.
- Your understanding will be put to use or tested in the lab.
- For programming practice:
 - You are recommended to install required software in your system.
 - The compiler is also accessible at <https://code.euclid-mu.in/>

Why this course?

- There is no prerequisite for this course.
- This is a prerequisite for everything else (Data Structures, Design Analysis and Algorithms, Advanced Algorithms)
 - Future courses
 - Placements
- Coding/Lots and lots of Practice
- Going Masters for a Master's Abroad

Course Syllabus

- This course is about programming concepts rather than a particular programming language.
- We will learn these programming concepts using C programming.
- The topics covered in this course are mentioned in the upcoming slides.

Course Objectives CS1001-Introduction to Computing (L-T-P-C: 3-0-2-4)

- This course is designed to develop a strong foundation in computers and programming through fundamental problem-solving techniques.
- It aims to nurture logical thinking in students using a learning-by-doing approach with the C programming language.
- The course includes a hands-on lab component to provide practical experience in implementing core programming concepts and solving real-world problems using the C programming language.

Course Syllabus

M1 [6 Hours]: Introduction to Computers and Programming

- Introduction to C, Structure of a C program, low-level vs. high-level languages.
- Number System: Binary arithmetic, Conversion from one base to another, Representation of negative and floating-point numbers, IEEE standards, and history.

M2 [6 Hours]: C Fundamentals & Basic Syntax

- Overview of C, Variables, Data types, and Constants.
- Operators & Expressions (Arithmetic, Relational, Logical, Bitwise).
- Basic I/O (other than file IO).

Course Syllabus

M3 [8 Hours]: Control Flow & Arrays

- Decision making and Branching: Conditions-if-else constructs, switch case, Ternary conditional statements.
- Decision making and looping: for loop, while loop, do-while, nested loops, break, continue.
- Arrays: 1D (one dimension) arrays, Multidimensional arrays, Character arrays, and Strings.

M4 [8 Hours]: User-Defined Functions

- Functions: Definition of function in C, call by value, functions with arrays.
- Recursive functions: Types of recursions - direct recursion, indirect recursion, Tail recursion.

Course Syllabus

M5 [8 Hours]: Pointers and Composite Data Types

- Pointers: Basics, pointer arithmetic, pointers vs. arrays.
- Dynamic Memory: malloc, realloc, calloc, free.
- Structures & Unions: Definition, nesting, pointers to structs, array of structures.
- Passing Variables: Call by value, call by reference, passing arrays and pointers.

M6 [6 Hours]: File Handling & Advanced Topics

- File Handling: File Opening Modes, fopen(), fread(), fwrite(), fclose().
- Advanced Topics: Command line arguments, debugging (gdb, assert()).

Textbook:

- T1: BW Kernighan, DM Ritchie, *The C Programming Language*, 2nd Ed, India: Prentice Hall, 1988.
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Reference Books:

R1: RG Dromey, *How to Solve it by Computer*, 1st Ed, Pearson, 2006.

R2: Yaswanth Kanethkar, *Let us C*, BPB Publications, 16th Ed, 2017.

R3: E Balaguruswamy, *Programming in ANSI C*, McGraw-Hill, 8th Ed, 2019.

R4: MD Harvey and PJ Deitel, *C: how to program*, Prentice-Hall, Inc., 1992.

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LAB ASSIGNMENTS:

- All Lab assignments will be done in a Unix-like shell environment.
- Each lab participation is mandatory for evaluation and award marks.
- **Make-up Policy:** It is as per the university norms.
- **Course Notices** will be communicated either by email or via Euclid.
- **Academic Honesty and Integrity Policy:** Academic honesty and integrity are to be maintained by all students throughout the semester, and any type of academic dishonesty is not acceptable.

Evaluation Criteria

Evaluation parameter	Weightage
Minor- I	15 %
Minor-2	15 %
End Semester	40 %
Lab Evaluation	Weekly Lab Assessment - 10 %
	Lab Exams – 20 %

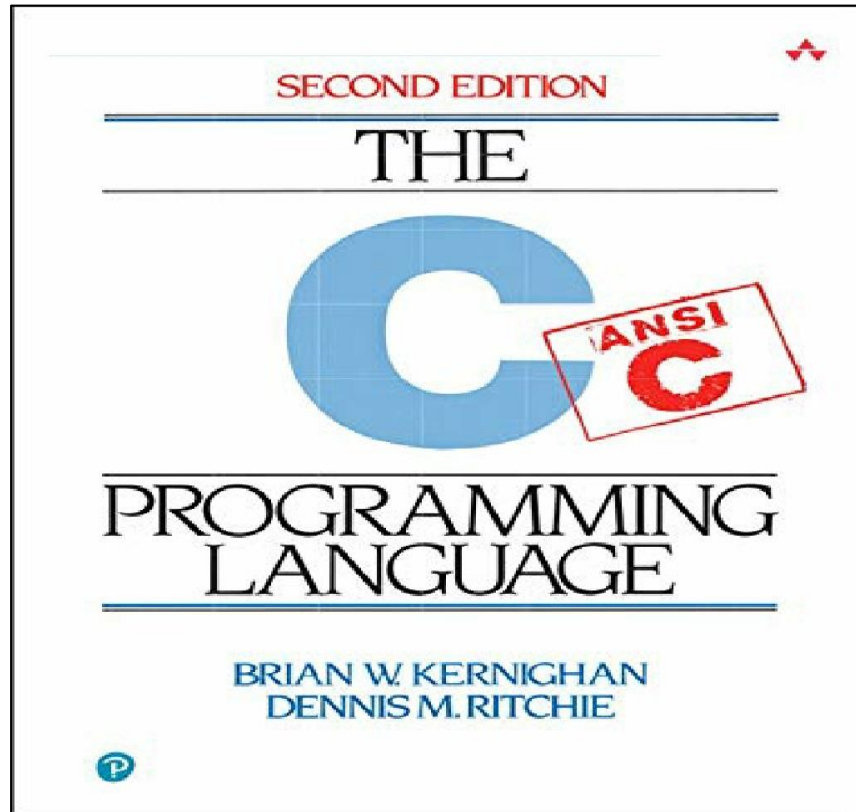
- Attendance is mandatory. Only QR code attendance will be considered.
- Regular lab sessions will be conducted.

Cheating Policy

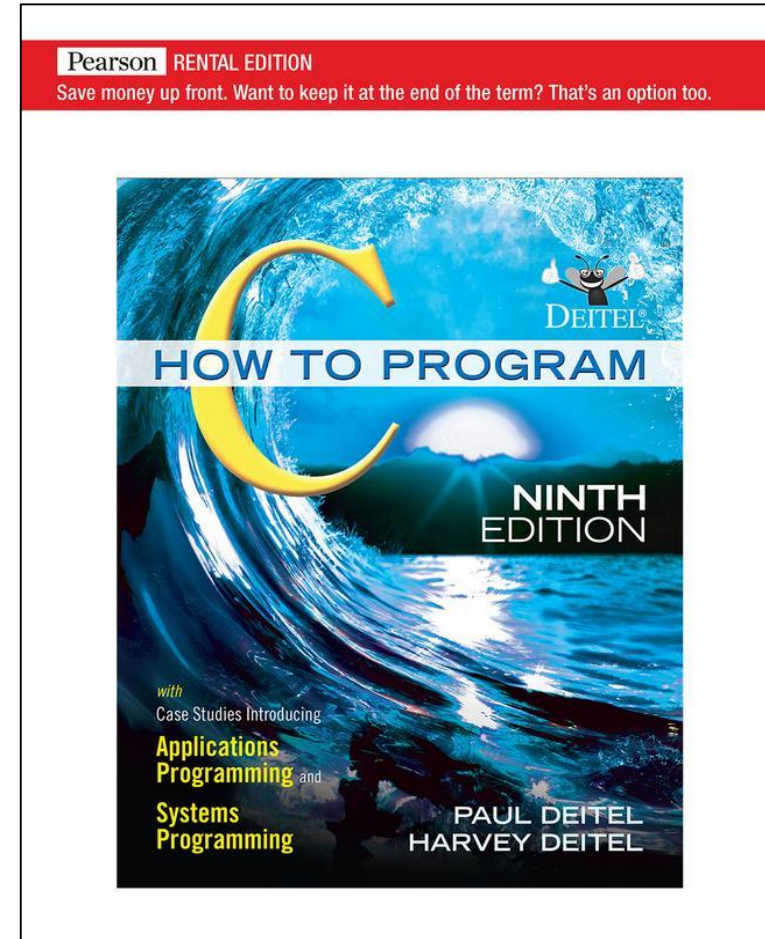
<https://www.mahindrauniversity.edu.in/sites/default/files/2022/Annexure-XVI-Unfair-Means.pdf>

- Lab Exams, minor, major exams are to be completed individually.
 - you must not share any code or other written material,
 - everything you turn in must be your own code/solution.
- If you find material in published book/literature (e.g., on the Web) that is helpful in solving a problem, you must cite it and explain the answer in your own words.
- Cheating in any exam is grave misconduct and a punishable offense. Plagiarism will thoroughly be checked for all submissions.
- We discourage students from adopting Cheating and encourage them to embrace academic honesty. Refer to the policy on Academic Unfair means.

Reference books/ materials



“The C Programming Language” by Brian Kernighan and Dennis Ritchie



“C How to Program” by Paul Deitel and Harvey Deitel

My Office hours

- Wednesday- 3:30-4:30 pm

Location- IT2 block, CSE Dept., 1st floor, cabin 33

- Make use of Office hours (Don't spam through individual emails!!!)
 - Instructors and TAs (Always for help!!!).
 - Don't come without a scheduled slot except during office hours.
 - Don't want to make you wait (in case of another meeting)!!

Therefore, a Calendar invite is a must even in case of in-person meetings!!

- **Course resources and materials link will be shared from time to time.**

